

ROS 2 - Motion Planning & Collision Objects

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Moveit Planners

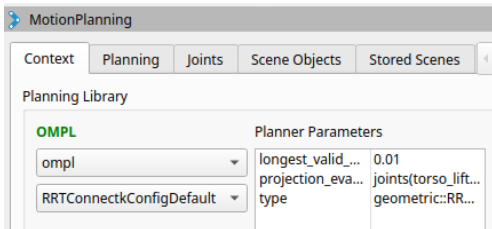
Several motion planners are available in Moveit2, based on the OMPL library.

- RRTConnectkConfigDefault (Bidirectional RRT)
- PRMkConfigDefault (Probabilistic Roadmap)
- TrajOptDefault (Trajectory Optimization for Motion Planning)
- ...

Change Moveit2 planner in Pymoveit2:

```
moveit2.planner_id = "RRTConnectkConfigDefault"
```

Rviz:



Collision Objects in Moveit2

By default, Moveit2 does not consider any collision object in the environment (except for the robot itself).

Pymoveit2 (check also Lab2) provides simple ways to add collision objects to the environment:

```
moveit2.add_collision_box(id, position, quat_xyzw, size)

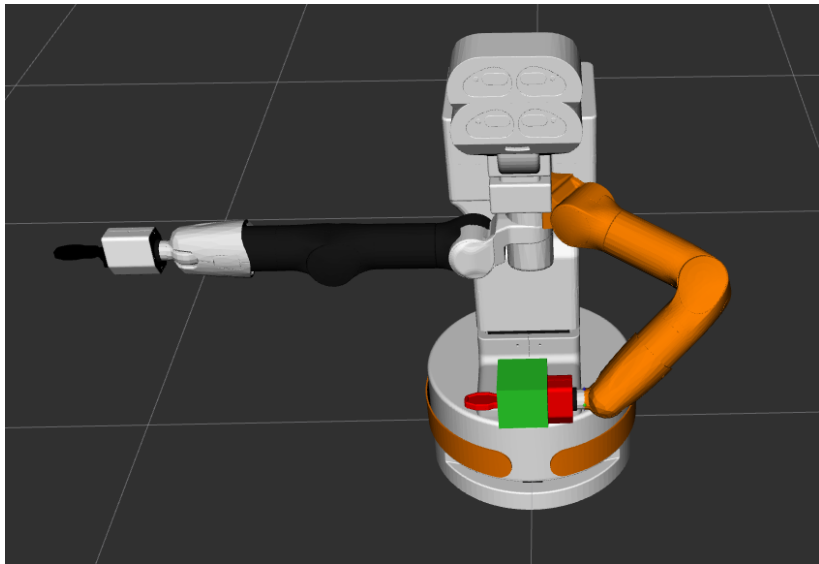
moveit2.add_collision_sphere(id=obj_id, position,
                             quat_xyzw, radius)

moveit2.add_collision_cylinder(id, position, quat_xyzw,
                               radius, height)
```

and to remove them:

```
moveit2.clear_all_collision_objects()
```

Example of Box Collision Objects



Modify the code of Lab2 to add/remove a collision object in the environment.